

‘A Fine Balance’: Exploring the Interplay of Reproductive Strategies and Growth Dynamics in Trees

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Masting is the phenomenon of synchronized and highly variable seed production within plant populations. It plays a critical role in ecosystem-level function, influencing both plant regeneration and the population dynamics of seed predators. Despite its importance, a mechanistic understanding of masting remains uncertain. Models trying to understand the phenomenon have been challenged by complex temporal dependencies and the difficulty of integrating environmental stochasticity with internal resource dynamics. While trade-offs between vegetative growth and reproduction are fundamental to tree life-history theory, they are often obscured by environmental stochasticity and the inherently unobservable nature of internal resource pools. Here we develop a new hierarchical Bayesian generative model to test one of many masting hypotheses—resource matching, which believes in that a fixed portion of resource is allocated to reproduction annually.

The model explicitly treats annual resource allocation as a latent process, proposing an unobserved total resource pool which is partitioned into two separate pools for growth and reproduction. By coupling two distinct data: annual tree-ring widths as a proxy for growth and annual seed production records as a proxy for reproduction collected from Mount Rainier National Park, the model reconstructs hidden resource pools from observed ecological data using allometric equations. By linking the allocation parameter and masting occurrence, we can better understand whether there exists a trade-off in the context of masting.

We discuss computational challenges and model degeneracies that arise because the allocation fraction and the total resource pool are mathematically coupled, leading to potential identifiability issues. Furthermore, we address the difficulty of linking latent allocation parameter to specific masting years when heavily constrained by the informativeness of the available ecological data. We utilize hierarchical structures to pool information across individuals, helping to estimate the latent state transitions. This model addresses the use of Bayesian modeling and latent variable frameworks in deciphering unobservable physiological processes from observable ecological data.